Q5		
(a)(i)	The component of velocity perpendicular to the magnetic field results in a magnetic force on the electron, which is always perpendicular to the motion. Hence, the electron rotates in a circular path. The component of velocity parallel to the magnetic field remains constant. Hence, the electron travels at constant velocity along the direction of magnetic field. The combination of both components of velocity of the electron results in a helical path.	
(a)(ii)	$F_B = Bqv_\perp = Bqv\sin 20°$ $v = \dfrac{F_B}{Bq\sin 20°} = \dfrac{4.3\times10^{-14}}{0.088(1.60\times10^{-19})\sin 20°}$ $v = 8.93\times10^{6}$ m s^{-1}	
(a)(iii)	Magnetic force provides centripetal force to the electron's component of velocity perpendicular to the magnetic field. Find period T: $F_B = Bqv_\perp = \dfrac{mv_\perp^2}{r} \Rightarrow v_\perp = \dfrac{Bqr}{m}$ $v_\perp = \omega r = \dfrac{2\pi}{T}r$ $T = \dfrac{2\pi r}{v_\perp} = \dfrac{2\pi rm}{Bqr} = \dfrac{2\pi m}{Bq}$ Pitch:` $p = v_{\parallel}T$ $= v\cos 20°\left(\dfrac{2\pi m}{Bq}\right)$ $= \dfrac{(8.93\times10^{6})\cos 20°(2\pi)(9.11\times10^{-31})}{(0.088)(1.60\times10^{-19})}$ $= 3.4\times10^{-3}$ m	
(b)	The pair of forces generate a torque. The maximum output torque is *F* x *d* $\text{Torque} = \tau = Fd = (NBIl)d = NBIA$ $B = \dfrac{\tau}{NIA} = \dfrac{395}{(1200)(96)(6.1\times10^{-3})}$ $B = 0.562$ T	